

Sentiment Analysis on Social Media Data

Machine Learning Capstone Project Report

Author: Arjun Mehta | Roll No: 21CS045 | B.Tech CSE — 4th Year

Institution: National Institute of Technology, Trichy | Date: November 2024

Abstract

This project implements a sentiment analysis pipeline on Twitter/X social media data using Natural Language Processing (NLP) and deep learning techniques. The model achieves 89.3% accuracy on a balanced dataset of 50,000 tweets, outperforming the baseline VADER lexicon approach by 12.4 percentage points.

Technologies Used

Python · TensorFlow · Keras · BERT (bert-base-uncased) · Hugging Face Transformers · NLTK · SpaCy · Pandas · NumPy · Matplotlib · Scikit-learn · Twitter API v2 · Google Colab

Methodology

1. Data Collection: Scraped 75,000 tweets using Twitter API v2 across 5 categories (positive, negative, neutral, sarcastic, mixed).
2. Preprocessing: Tokenization, stop-word removal, lemmatization using SpaCy.
3. Model: Fine-tuned BERT transformer with custom classification head (3-class output).
4. Training: AdamW optimizer, learning rate $2e-5$, batch size 32, 5 epochs on T4 GPU.
5. Evaluation: Accuracy 89.3%, F1-score 0.917, AUC-ROC 0.94.

Results

Final model accuracy: 89.3% on held-out test set.

The project was evaluated as part of the 6th semester Machine Learning course under Dr. Priya Sundarajan with a grade of **A+ (9.5/10)**.